END-OF-CHAPTER EXERCISES — COMPLETE SOLUTIONS

Ganita Manjari | Grade 9 | Chapter: Quadrilaterals (Pages 75–83)

Question 1

Using a fact about parallelograms, show how to tile the plane using any given triangle. (Hint: Can you use the parallelogram tiling in the introduction?)

SOLUTION:

1. Take two identical (congruent) copies of any given triangle ΔABC.
2. Rotate one copy by $180°$ and place it adjacent to the other along a common side (e.g., side AC).
3. This combined 2-triangle shape forms a **parallelogram**.
4. Since any parallelogram can seamlessly tile the 2D plane through horizontal and vertical shifts (translations), repeating this combined parallelogram block covers the entire plane without gaps or overlaps.

Question 2

Mark the midpoint of the line drawn on the paper (see Fig. 12.33), given that the horizontal lines are equally spaced. Justify your answer.

SOLUTION:

1. Count the total number of equal horizontal parallel spaces intersected by the line segment. Suppose the segment spans across N equal parallel strips.
2. By the **Intercept Theorem / Equal Intercepts Property**: If three or more parallel lines make equal intercepts on one transversal, they make equal intercepts on any other transversal.
3. Identify the middle parallel line corresponding to $N / 2$ strips.
4. The intersection point of this middle horizontal line with the given slanted line segment is the exact **midpoint** of the line segment.

Question 3

You know that the sum of angles of a quadrilateral is $360°$, even for a non-convex quadrilateral. Now consider a self-intersecting quadrilateral $ABCD$, where AB and CD intersect at point E. Show that $\angle A + \angle B + \angle C + \angle D < 360°$. Can you construct $ABCD$ such that $\angle A + \angle B + \angle C + \angle D = 2°$?

SOLUTION:

1. The self-intersecting quadrilateral forms two triangles, ΔAED and ΔBEC, crossing at vertex E.
2. In ΔAED: $\angle A + \angle D + \angle AED = 180° \Rightarrow \angle A + \angle D = 180° - \angle AED$.
3. In ΔBEC: $\angle B + \angle C + \angle BEC = 180° \Rightarrow \angle B + \angle C = 180° - \angle BEC$.
4. Since $\angle AED$ and $\angle BEC$ are vertically opposite angles, $\angle AED = \angle BEC = \theta$ (where $\theta > 0°$).
5. Adding the four vertex angles:

   > $\angle A + \angle B + \angle C + \angle D = (180° - \theta) + (180° - \theta) = 360° - 2\theta < 360°$

6. **Constructing total sum $= 2°$:** Yes! Setting $360° - 2\theta = 2° \Rightarrow 2\theta = 358° \Rightarrow \theta = 179°$. By making lines AB and CD intersect at a very obtuse angle ($\theta = 179°$), the sum of vertex angles becomes exactly $2°$.

Question 4

In a parallelogram $ABCD$, two points P and Q are taken on diagonal BD such that $DP = BQ$ (see Fig. 12.34). Show that $APCQ$ is a parallelogram.

SOLUTION:

1. Let diagonals AC and BD intersect at point O.
2. Since $ABCD$ is a parallelogram, its diagonals bisect each other $\Rightarrow OA = OC$ and $OB = OD$.
3. Given $DP = BQ$. Subtracting these equal lengths from $OD = OB$:

   $$OD - DP = OB - BQ \Rightarrow OP = OQ$$

4. In quadrilateral $APCQ$, the diagonals are AC and PQ, intersecting at O. Since $OA = OC$ and $OP = OQ$, the diagonals of $APCQ$ bisect each other.
5. Therefore, **$APCQ$ is a parallelogram**.

Question 5

A right-triangle shaped cutout of paper is folded such that point A touches point B (Fig. 12.35). Show that the crease line can be used to find the midpoint of not only AB but also that of AC.

SOLUTION:

1. Folding point A onto point B creates a crease line that is the **perpendicular bisector** of side AB. This crease intersects AB at its midpoint M.
2. Since ΔABC is right-angled at B ($AB \perp BC$), the crease line (perpendicular to AB) is **parallel to side** BC.
3. Let the crease line intersect hypotenuse AC at point N.
4. By the **Converse of Midpoint Theorem** in ΔABC: A line through midpoint M of AB parallel to BC bisects the third side AC.
5. Hence, point N is the **midpoint of** AC.

Question 6 (Lee Sallows Dissection)

(i) Cut along medians AP, BQ, CR of ΔABC (centroid M) to get 6 triangles. Reassemble into 3 congruent triangles. Prove side lengths.
(ii) Repeat procedure to get 9 triangles with sides $AB/3$, $BC/3$, $AC/3$.
(iii) Reassembling a triangle into 2 congruent triangles using median AP.

SOLUTION:

1. **(i) 3 Congruent Triangles:** Pair the 6 smaller centroid-divided triangles along matching segments of medians (MP, MQ, MR). Re-joining two adjacent small pieces around each vertex creates 3 congruent triangles, each having sides equal to $\frac{2}{3}$ of the original medians. Their side lengths are proportional to $AB/2$, $BC/2$, $AC/2$.
2. **(ii) 9 Triangles:** Applying the same centroid median dissection recursively to each of the 3 newly formed triangles subdivides the whole figure into 9 smaller triangles, each having sides scaled down by $\frac{1}{3}$: $AB/3$, $BC/3$, $AC/3$.
3. **(iii) 2 Congruent Triangles:** Cut along median AP of ΔABC to split it into ΔABP and ΔACP (equal areas). Then cut each along their respective inner medians to rearrange into 2 congruent halves.

Question 7

In trapezium $ABCD$ ($AB \parallel DC$), E is midpoint of AD. A line through E intersects BC at F.
(i) If $EF \parallel AB$, show F is midpoint of BC and $EF = (AB + CD)/2$.
(ii) If F is midpoint of BC, show $EF \parallel AB$.

SOLUTION:

1. **(i) Proof:** Join diagonal BD, intersecting EF at M.
   - In ΔDAB, E is midpoint of AD and $EM \parallel AB \Rightarrow M$ is midpoint of BD, and $EM = \frac{1}{2} AB$.
   - In ΔBCD, M is midpoint of BD and $MF \parallel DC \Rightarrow F$ is midpoint of BC, and $MF = \frac{1}{2} CD$.
   - Adding lengths: $EF = EM + MF = \frac{1}{2} AB + \frac{1}{2} CD = (AB + CD) / 2$.
2. **(ii) Proof:** Join midpoints E (of AD) and M (of BD) $\Rightarrow EM \parallel AB$. Join M and F (of BC) $\Rightarrow MF \parallel CD \parallel AB$. By Euclid's Parallel Postulate, E, M, F lie on a single line $\Rightarrow EF \parallel AB$. (Direct method via diagonal midpoint M is simpler).

Question 8

Diagonals AC, BD of parallelogram $ABCD$ intersect at O. A line through O meets AB at P and CD at Q. Show O is midpoint of PQ.

SOLUTION:

1. In ΔAOP and ΔCOQ:
   - $OA = OC$ (Diagonals of parallelogram bisect each other)
   - $\angle AOP = \angle COQ$ (Vertically opposite angles)
   - $\angle PAO = \angle QCO$ (Alternate interior angles, since $AB \parallel CD$)
2. By **ASA Congruence Criteria**: $\Delta AOP \cong \Delta COQ$.
3. By CPCTC: $OP = OQ$. Hence, O is the midpoint of PQ.

Question 9

Trapezium $ABCD$ has parallel sides $AD = 3$ cm and $BC = 5$ cm. E, F are midpoints of non-parallel sides. Find ratio of areas of $AEFD$ and $EBCF$.

SOLUTION:

1. By Trapezium Midline Theorem (Q7): $EF = (AD + BC) / 2 = (3 + 5) / 2 = 4$ cm.
2. The line EF divides the height h of trapezium $ABCD$ into two equal heights $h_1 = h_2 = h/2$.
3. Area of upper 4-gon $AEFD = \frac{1}{2} \times (AD + EF) \times (h/2) = \frac{1}{2} \times (3 + 4) \times (h/2) = (7/4)\ h$.
4. Area of lower 4-gon $EBCF = \frac{1}{2} \times (EF + BC) \times (h/2) = \frac{1}{2} \times (4 + 5) \times (h/2) = (9/4)\ h$.
5. Ratio of Areas = $(7/4\ h) / (9/4\ h) = 7 : 9$.

Question 10

4 points A, B, C, D (no three collinear).
(i) How many different quadrilaterals? (ii) How many self-intersecting? How many convex?

SOLUTION:

1. **(i) Total Quadrilaterals:** For 4 points, there are 3 distinct circular orderings to connect them into a closed 4-gon: $3! / 2 = 3$ different quadrilaterals.
2. **(ii) Breakdown by Type:**
   - If 1 point lies inside the triangle formed by the other 3: **0 convex, 1 non-convex (dart), 2 self-intersecting**.
   - If all 4 points form a convex boundary: **1 convex, 0 non-convex, 2 self-intersecting**.

Question 11

In ΔABC, line through P on AB parallel to BC meets AC at Q. $AP = 1$, $AQ = \sqrt{5}$.
(i) If $PB = 2$, find QC. (ii) If $PB = \frac{1}{3}$, find QC. (iii) If $PB = \frac{2}{3}$, find QC. (iv) Show $AP/PB = AQ/QC$.

SOLUTION:

1. By **Basic Proportionality Theorem (Thales' Theorem)**: Since $PQ \parallel BC$, $AP / PB = AQ / QC$.
2. **(i)** $PB = 2$: $1 / 2 = \sqrt{5} / QC \Rightarrow QC = 2\sqrt{5}$.
3. **(ii)** $PB = \frac{1}{3}$: $1 / (1/3) = \sqrt{5} / QC \Rightarrow 3 = \sqrt{5} / QC \Rightarrow QC = \sqrt{5} / 3$.
4. **(iii)** $PB = \frac{2}{3}$: $1 / (2/3) = \sqrt{5} / QC \Rightarrow 3/2 = \sqrt{5} / QC \Rightarrow QC = 2\sqrt{5} / 3$.
5. **(iv) General Rational Proof:** Dividing AP and PB into equal unit grid steps establishes $AP / PB = AQ / QC$ for all rational ratios.

Question 12

(i) Parallelogram $ABCD$, M, N midpoints of AB, CD. Show DM, BN trisect diagonal AC.
(ii) Trisection methods for any segment PQ.

SOLUTION:

1. **(i) Proof:** $AM = NC$ and $AM \parallel NC \Rightarrow AMCN$ is a parallelogram $\Rightarrow AN \parallel MC$.
   - In ΔDCP (where DM intersects AC at P, BN intersects AC at Q): N is midpoint of DC and $NQ \parallel DP \Rightarrow Q$ is midpoint of $PC \Rightarrow PQ = QC$.
   - In ΔABQ: M is midpoint of AB and $MP \parallel BQ \Rightarrow P$ is midpoint of $AQ \Rightarrow AP = PQ$.
   - Thus, $AP = PQ = QC$, trisecting diagonal AC.
2. **(ii) Other Trisection Methods:**
   - **Thales Parallel Line Method:** Draw a ray from P, mark 3 equal arcs, connect end to Q, and draw parallel lines.
   - **Centroid Method:** Construct a triangle on PQ as median; the centroid divides medians in $2:1$ ratio.

Question 13

Midpoint theorem for non-convex / self-intersecting quadrilaterals.

SOLUTION:

1. **(i) Validity:** The Midpoint Theorem depends purely on vector additions / triangle midpoint properties along diagonals ($PQ = \frac{1}{2} AC = SR$), which hold independently of whether the shape is convex or non-convex.
2. **(ii) Exception:** When all 4 midpoints become collinear, the Varignon parallelogram collapses into a degenerate single line segment.
3. **(iii) Non-planar (3D) Quadrilaterals:** The theorem remains true even in 3D space! Midpoints of any spatial 4-gon always form a planar parallelogram.

Question 14

Converse question to Varignon Theorem.

SOLUTION:

If midpoints P, Q, R, S form a parallelogram, it does not uniquely determine a single quadrilateral because multiple original quadrilaterals share the exact same Varignon parallelogram (as proved in Ex 12.3 Q5).

Question 15

Midpoint Theorem proof extension: Extend PQ beyond Q to point S such that $PQ = QS$.

SOLUTION:

1. In ΔAPQ and ΔCSQ: $AQ = QC$, $PQ = QS$, $\angle AQP = \angle CQS$ (vertically opposite).
2. By **SAS Criteria**: $\Delta APQ \cong \Delta CSQ$.
3. Thus $AP = CS$ and $\angle PAQ = \angle SCQ \Rightarrow CS \parallel AB$.
4. Since $AP = PB$, we get $PB = CS$ and $PB \parallel CS \Rightarrow PBCS$ is a parallelogram.
5. Hence $PS \parallel BC$ and $PS = BC \Rightarrow PQ = \frac{1}{2} BC$ and $PQ \parallel BC$.

Question 16

Converses of Rhombus / Rectangle / Square properties.

SOLUTION:

- **Rectangle:** A parallelogram with equal diagonals is a rectangle (True).
- **Rhombus:** A parallelogram with perpendicular diagonals is a rhombus (True).
- **Square:** A parallelogram with equal AND perpendicular diagonals is a square (True).

Question 17

Non-planar quadrilateral angle sum. Show $\angle A + \angle B + \angle C + \angle D < 360°$. Can total sum = 2°?

SOLUTION:

1. Bending a planar quadrilateral along diagonal AC like a hinge into 3D space decreases the effective angles at vertices B and D relative to the 3D projection.
2. Thus, the sum of interior angles of any non-planar quadrilateral is strictly **less than 360°**.
3. By folding the two triangular flaps ABC and ADC almost flat against each other (nearly 180° fold), the total sum of angles can be made as small as $2°$.

Question 18

Tiling the plane using a 4-gon (3rd method - diagonal reflection/translation).

SOLUTION:

Matching two copies of quadrilateral ***SOME*** along diagonal ***EO*** creates a symmetric composite 6-gon block. By translating this composite block along the grid direction defined by the diagonals, the entire 2D plane is covered seamlessly without gaps or overlaps.

Question 19

Side length conditions for 4-gons.
(i) Triangle inequality recall.
(ii) Can 4-gon have sides 2, 5, 11, and 100? 10? 1?
(iii) General condition.

SOLUTION:

1. **(i) Triangle Inequality:** Sum of any two sides must be greater than the third side ($a + b > c$).
2. **(ii) Quadrilateral Side 100:** No, because $2 + 5 + 11 = 18 < 100$ (cannot close).
   - **Side 10:** Yes, because $10 < 2 + 5 + 11 = 18$.
   - **Side 1:** Yes, because $11 < 2 + 5 + 1 = 8$ is false? Wait! Max side is 11: $11 < 2 + 5 + 1 = 8$ (No, 11 is greater than 8, so side 1 cannot form a closed 4-gon!).
3. **(iii) General Rule (Polygon Inequality):** For any polygon, the longest side must be strictly less than the sum of all other sides:

   > $d < a + b + c$

Question 20

Counting diagonals of an *n*-gon.

SOLUTION:

1. **Values for small *n*:**
   - 3-gon: 0 diagonals
   - 4-gon: 2 diagonals
   - 5-gon: 5 diagonals
   - 6-gon: 9 diagonals
   - 7-gon: 14 diagonals
   - 8-gon: 20 diagonals
2. **General Formula Derivation:** From each of the ***n*** vertices, we can draw lines to ***(n - 3)*** non-adjacent vertices. Since each diagonal connects 2 vertices, dividing by 2 avoids double counting:

   > ***Number of Diagonals** = [n × (n − 3)] / 2*

Question 21

Sum of angles of an *n*-gon.

SOLUTION:

1. Connecting one vertex to all other non-adjacent vertices divides an ***n***-gon into ***(n - 2)*** non-overlapping triangles.
2. Since the angle sum of each triangle is ***180°***:

   > ***Sum of Interior Angles** = (n − 2) × 180°*

3. Values: ***n=3 ⇒ 180°***, ***n=4 ⇒ 360°***, ***n=5 ⇒ 540°***, ***n=6 ⇒ 720°***.

Question 22

Multiple converses to Midpoint Theorem.

SOLUTION:

The Midpoint Theorem has 2 assumptions (P is midpoint, Q is midpoint) and 2 conclusions (***PQ ∥ BC***, ***PQ = ½ BC***). Swapping any one or both conclusions with the assumptions yields valid converse statements, most notably **Theorem 7 (Converse of Midpoint Theorem)**: A line through the midpoint of one side parallel to the base bisects the second side.

Question 23

Validity of tiling methods for any 4-gon.

SOLUTION:

Since the sum of interior angles of any quadrilateral is $360°$, placing 4 copies of any quadrilateral around every vertex such that all 4 distinct angles meet ensures that the vertex angle sum is exactly $360°$, guaranteeing gapless and overlap-free tiling of the entire 2D plane.

Question 24

What fraction of the square is shaded? (Fig. 12.43)

SOLUTION:

1. Let the square have side length $s = 1$ unit (Total Area = 1).
2. The lines connecting vertices to midpoints of opposite sides form a central tilted square (shaded region).
3. By standard geometric dissection / coordinate geometry for this classic midpoint square construction:

> ***Area of Shaded Region = 1 / 5 = 20% of the total square area***